

Dhia Problem 3 — Inclined Sheet (New Formula): Results Report

Dusty Al_2O_3 -Cu/water hybrid nanofluid boundary layer over an inclined permeable stretching/shrinking sheet with velocity and thermal slip. Program: *Dhia_Problem3_Inclined_New_Formula.m*, tasks as specified in PROMPT.docx.

Method note. MATLAB is not installed on this machine, so the bvp4c program was ported 1:1 to Python (scipy.integrate.solve_bvp, a collocation method equivalent to bvp4c, RelTol = $1\text{e-}8$). Validation: (i) at the .m file's own default parameters both shipped initial guesses converge cleanly; (ii) Task 1 with $\sigma_1 = 0$ gives $f''(0) = -0.99977 \approx -1$, the exact classical Crane value; (iii) Gr = 1 with $X = \pi/2$ ($\cos X = 0$) exactly reproduces the Gr = 0.001 curve. Parameters not named in the PROMPT keep the .m file defaults: $\sigma_2 = 0.2$, Gr = 0.001, L = 1.

Task 1 — Effect of velocity slip σ_1

$\lambda = 1$, $\phi_1 = \phi_2 = \beta_v = \beta_T = s = X = 0$, Pr = 6.2, Ec = 1, m = 1, $\varepsilon = 1$. Values are domain-converged (verified up to $\eta_{\max} = 30$).

sigma1	f''(0)	F'(0)	-t'(0)	tp(0)
0.0	-0.999770355	0.000000	1.405296308	0.000000
0.1	-0.871889376	0.000000	1.487037403	0.000000
0.2	-0.776208327	0.000000	1.536456760	0.000000
0.5	-0.591067377	0.000000	1.597089337	0.000000
2.0	-0.283903629	0.000000	1.538354886	0.000000
5.0	-0.144784538	0.000000	1.369498066	0.000000

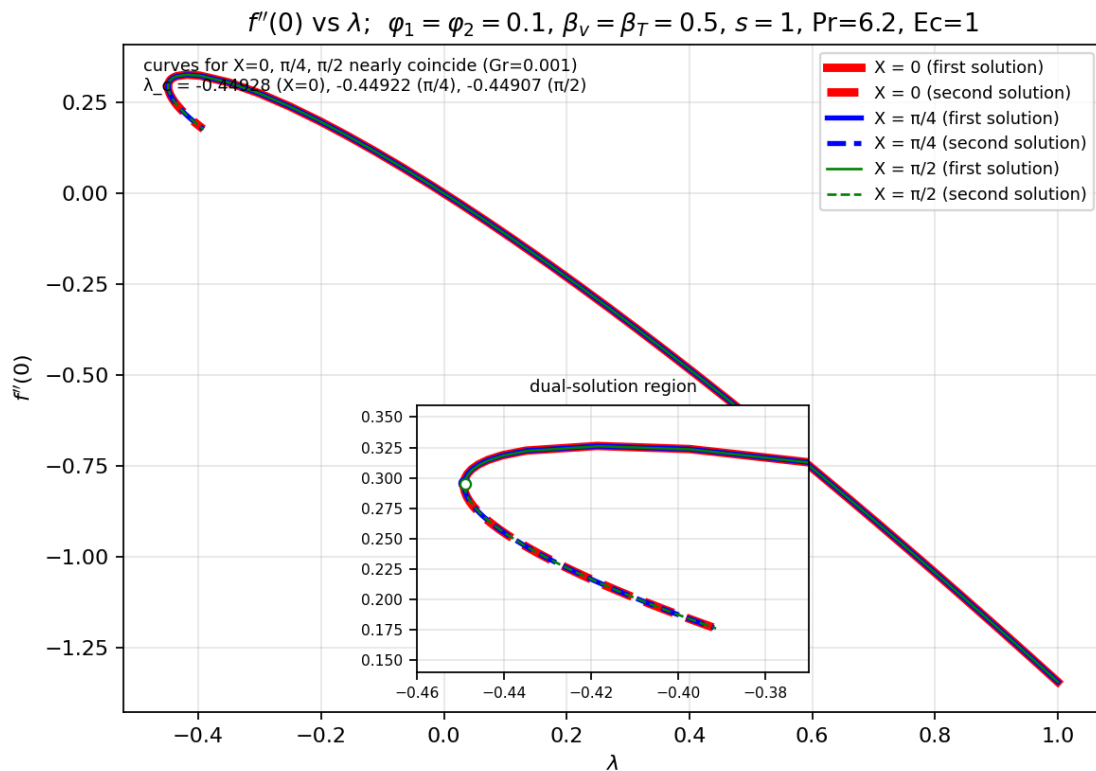
No dual solution exists for this case — $\lambda = 1$ is a stretching case (dual solutions arise for shrinking, $\lambda < 0$); both of the program's initial guesses (OdeInit1 #6 and OdeInit2 #7) converge to the same solution. With $\beta_v = \beta_T = 0$ the dust phase decouples completely, so $F'(0) = t_p'(0) = 0$ exactly. Note: at $\sigma_1 = 2.0$ an isolated mathematical second solution exists ($f''(0) = -0.4687$, $-t'(0) = 27.14$) but it has $t(0) = -4.4 < 0$ and cannot be continued to neighbouring σ_1 values — a thermal eigen-resonance, not a physical dual solution.

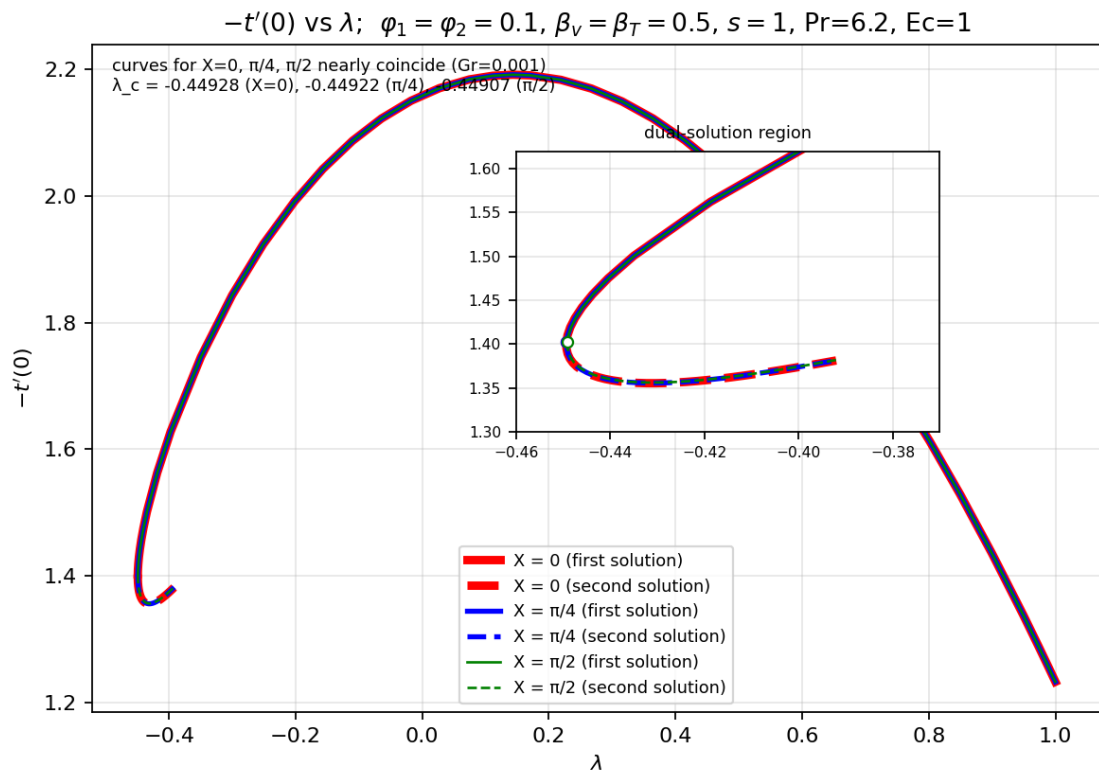
Task 2 — $f''(0)$ and $-t'(0)$ versus λ for $X = 0, \pi/4, \pi/2$

$\phi_1 = \phi_2 = 0.1$, $\beta_v = \beta_T = 0.5$, $s = 1$, $Pr = 6.2$, $Ec = 1$, $m = 1$, $\varepsilon = 1$ ($\sigma_1 = \sigma_2 = 0.2$, $Gr = 0.001$, $L = 1$). Solid line = first solution, dashed = second solution. Dual solutions exist between the critical point λ_c and $\lambda \approx -0.39$, beyond which the second solution undergoes thermal-resonance blow-up of $-t'(0)$ (an effect of the $Ec = 1$ viscous dissipation term).

X	lambda_c	$f''(0)$ at fold	$-t'(0)$ at fold
0	-0.449277	0.29	1.40
$\pi/4$	-0.449217	0.29	1.40
$\pi/2$	-0.449072	0.29	1.40

The three inclination curves nearly coincide: X enters the momentum equation only through $Gr \cdot A_g \cdot t \cdot \cos(X)$ and the program's $Gr = 0.001$ makes this term negligible. The supplementary $Gr = 1$ figures on the next pages show the inclination effect becoming visible when the mixed-convection parameter is $O(1)$.





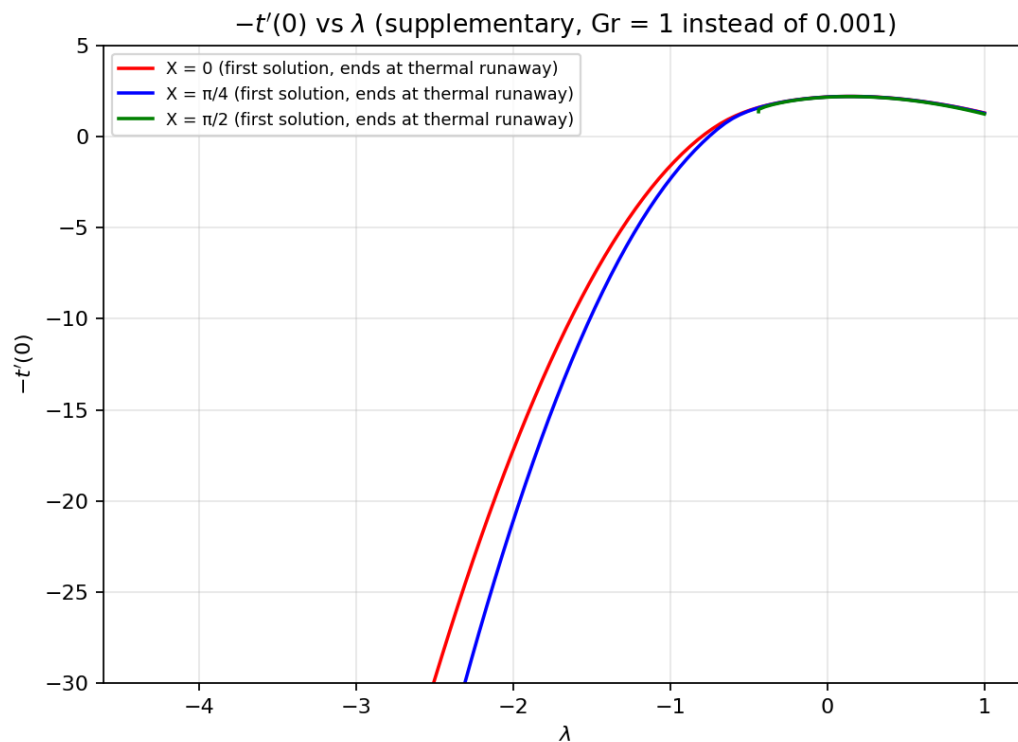
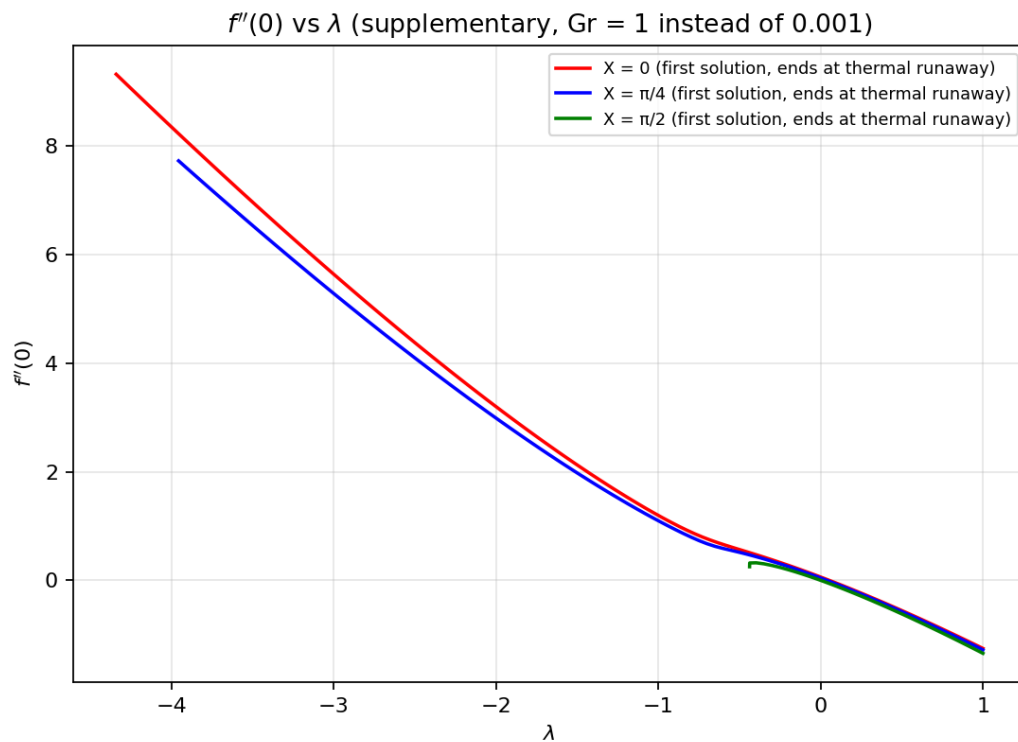
Selected curve data, $X = 0$ (full data in task2_curve_X0/Xpi4/Xpi2.csv; curves for $\pi/4$ and $\pi/2$ agree to the 3rd–4th decimal)

branch	lambda	$f''(0)$	$F'(0)$	$-t'(0)$	$tp(0)$
first	1.00000	-1.343982	0.119115	1.232226	0.093318
first	0.94917	-1.266513	0.115364	1.337950	0.089672
first	0.80156	-1.045911	0.103634	1.612268	0.079865
first	0.65588	-0.835111	0.090623	1.833904	0.071546
first	0.51152	-0.633788	0.075979	2.003451	0.064994
first	0.36447	-0.437497	0.058764	2.122407	0.060579
first	0.21909	-0.253532	0.038721	2.183494	0.059250
first	0.07410	-0.082038	0.014564	2.184085	0.062216
first	-0.06458	0.067977	-0.014309	2.122280	0.071170
first	-0.20158	0.197970	-0.051924	1.991529	0.090020
first	-0.34959	0.304626	-0.113964	1.745646	0.135239
first	-0.43494	0.322433	-0.185247	1.500670	0.205049
first	-0.44595	0.310939	-0.207084	1.441918	0.229876
first	-0.44928	0.295615	-0.224783	1.402071	0.251087
second	-0.44928	0.295615	-0.224783	1.402071	0.251087
second	-0.44789	0.280291	-0.237743	1.378546	0.267168
second	-0.44247	0.261136	-0.250193	1.362125	0.282956
second	-0.43424	0.241981	-0.259718	1.355687	0.295157
second	-0.42609	0.226657	-0.265650	1.356065	0.302752

second	-0.41424	0.207502	-0.271236	1.362092	0.309884
second	-0.40077	0.188347	-0.274978	1.372994	0.314778
second	-0.42511	0.224961	-0.266224	1.356366	0.303484

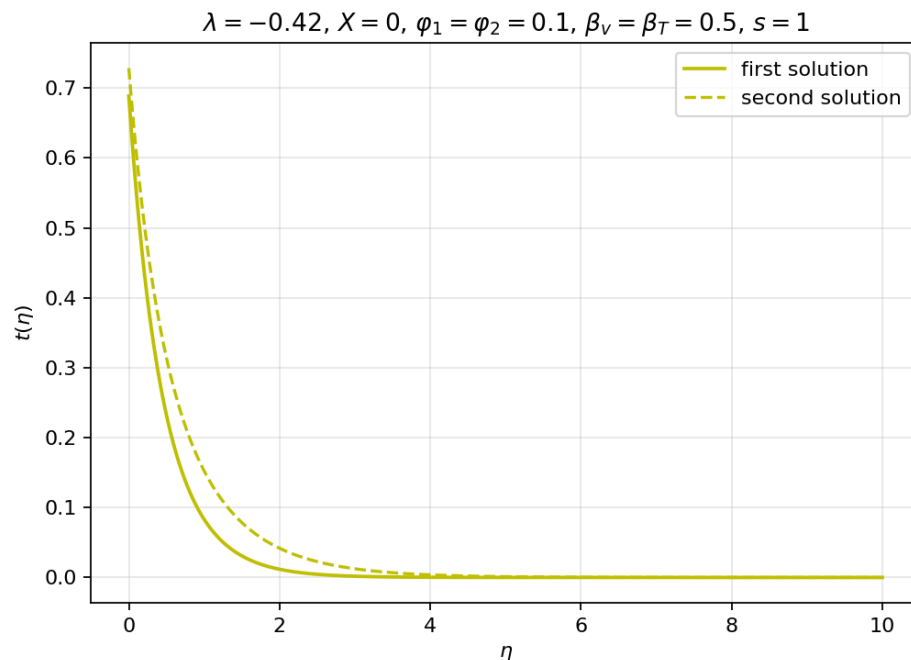
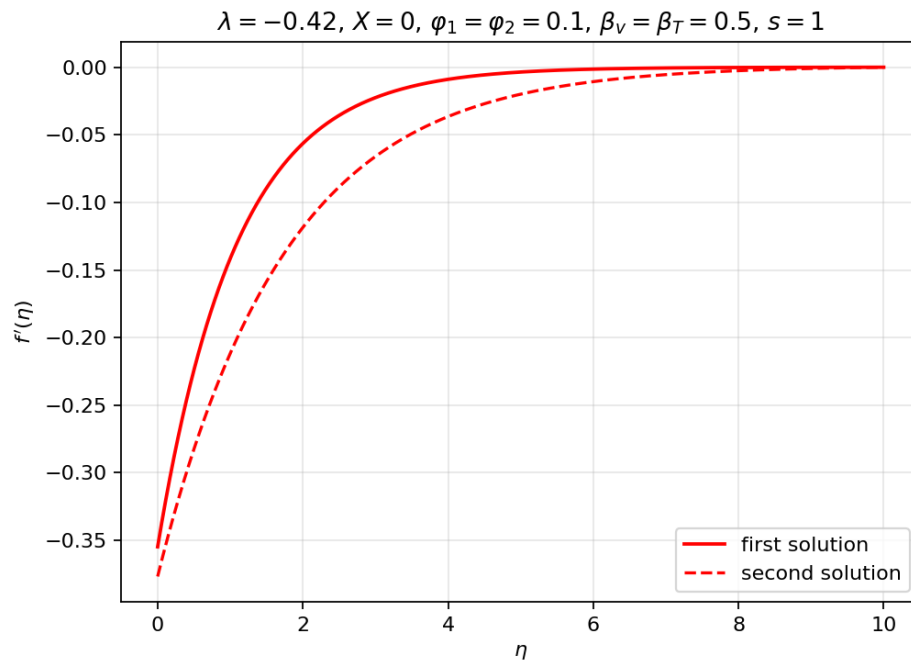
Task 2 (supplementary) — same sweep with $Gr = 1$

With $Gr = 1$ the inclination effect is clearly visible (buoyancy assists most at $X = 0$). For $X = 0$ and $\pi/4$ no fold is reached: the Ec-driven temperature growth ends the branch in thermal runaway near $\lambda \approx -4.3$ and -4.0 respectively. $X = \pi/2$ ($\cos X = 0$) reproduces the $Gr = 0.001$ result, fold and second branch included.

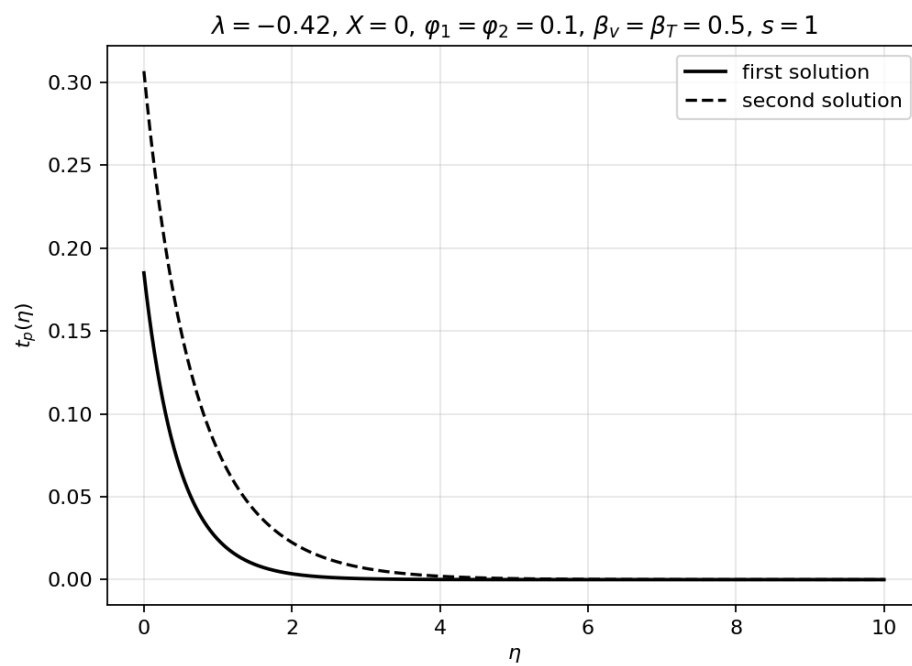
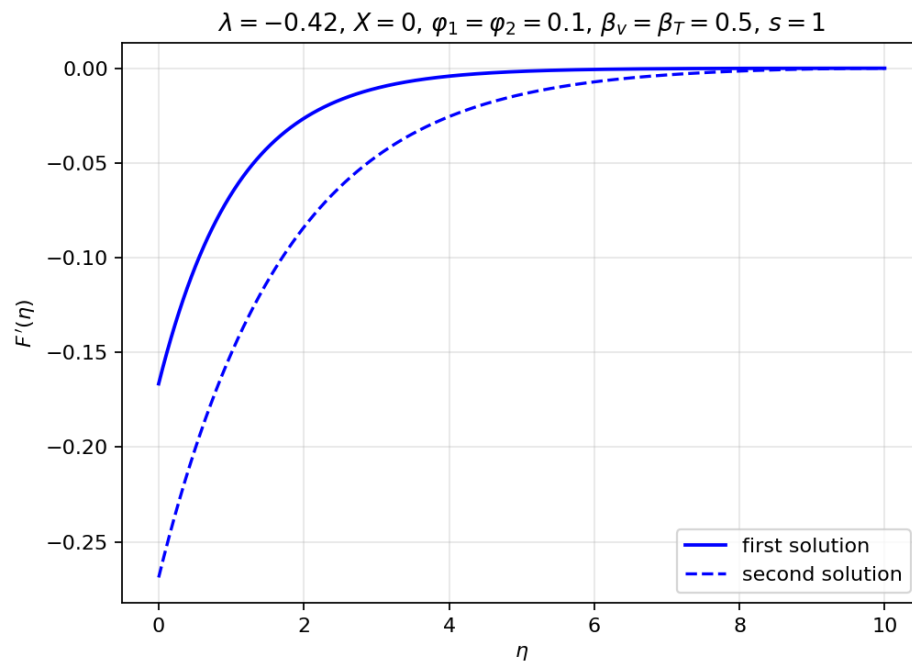


Task 3 — Dual velocity and temperature profiles ($\lambda = -0.42, X = 0$)

Profiles of both solutions at $\lambda = -0.42$ inside the dual region. First solution: $f''(0) = 0.326205$, $-t'(0) = 1.557746$; second solution: $f''(0) = 0.216472$, $-t'(0) = 1.358579$. All eight OdeBC boundary conditions are satisfied to $\sim 1e-17$ and every profile decays to zero far from the wall: $f(0) = s$, $f'(0) = \lambda + \sigma_1 f''(0)$, $t(0) = 1 + \sigma_2 t'(0)$, $f(\infty) = F(\infty) = t(\infty) = t_p(\infty) = 0$, $F(\infty) = f(\infty)$.



Dust-phase profiles



Task 3 profile data (sampled; full data in task3_profile_*.csv)

First solution

eta	f	f'	f''	F	F'	t	t'	tp
0.00	1.00000	-0.35476	0.32620	0.79535	-0.16661	0.68845	-1.55775	0.18510
0.50	0.85780	-0.22402	0.20595	0.72857	-0.10521	0.23194	-0.48807	0.06559
1.00	0.76801	-0.14147	0.13006	0.68640	-0.06643	0.08305	-0.16707	0.02418
1.50	0.71130	-0.08933	0.08213	0.65978	-0.04194	0.03085	-0.06027	0.00914
2.00	0.67549	-0.05641	0.05187	0.64297	-0.02648	0.01173	-0.02248	0.00352
2.50	0.65288	-0.03561	0.03276	0.63235	-0.01671	0.00453	-0.00857	0.00137
3.00	0.63861	-0.02248	0.02069	0.62566	-0.01054	0.00177	-0.00331	0.00054
3.50	0.62960	-0.01419	0.01307	0.62144	-0.00665	0.00069	-0.00129	0.00021
4.00	0.62392	-0.00895	0.00825	0.61878	-0.00419	0.00027	-0.00051	0.00008
4.50	0.62033	-0.00564	0.00521	0.61710	-0.00263	0.00011	-0.00020	0.00003
5.00	0.61808	-0.00355	0.00329	0.61605	-0.00165	0.00004	-0.00008	0.00001
5.50	0.61666	-0.00223	0.00208	0.61539	-0.00103	0.00002	-0.00003	0.00001
6.00	0.61577	-0.00140	0.00131	0.61498	-0.00064	0.00001	-0.00001	0.00000
6.50	0.61521	-0.00087	0.00083	0.61473	-0.00039	0.00000	-0.00000	0.00000
7.00	0.61487	-0.00054	0.00052	0.61458	-0.00024	0.00000	-0.00000	0.00000
7.50	0.61465	-0.00033	0.00033	0.61448	-0.00014	0.00000	-0.00000	0.00000
8.00	0.61453	-0.00019	0.00021	0.61443	-0.00008	0.00000	-0.00000	0.00000
8.50	0.61445	-0.00011	0.00013	0.61440	-0.00004	0.00000	-0.00000	0.00000
9.00	0.61441	-0.00006	0.00008	0.61439	-0.00002	0.00000	-0.00000	0.00000
9.50	0.61439	-0.00002	0.00005	0.61439	-0.00000	0.00000	-0.00000	0.00000
10.00	0.61439	-0.00000	0.00004	0.61439	0.00000	-0.00000	-0.00000	0.00000

Second solution

eta	f	f'	f''	F	F'	t	t'	tp
0.00	1.00000	-0.37671	0.21647	0.81190	-0.26886	0.72828	-1.35858	0.30685
0.50	0.83631	-0.28251	0.16283	0.69517	-0.20128	0.31386	-0.48279	0.15038
1.00	0.71361	-0.21168	0.12239	0.60782	-0.15056	0.15231	-0.20902	0.07779
1.50	0.62170	-0.15846	0.09193	0.54251	-0.11250	0.07853	-0.10100	0.04144
2.00	0.55294	-0.11850	0.06901	0.49373	-0.08396	0.04182	-0.05181	0.02245
2.50	0.50155	-0.08850	0.05178	0.45735	-0.06257	0.02268	-0.02748	0.01229
3.00	0.46319	-0.06601	0.03883	0.43027	-0.04654	0.01243	-0.01486	0.00677
3.50	0.43461	-0.04914	0.02911	0.41014	-0.03454	0.00685	-0.00813	0.00374
4.00	0.41335	-0.03649	0.02181	0.39523	-0.02555	0.00379	-0.00448	0.00207
4.50	0.39759	-0.02702	0.01634	0.38422	-0.01882	0.00211	-0.00248	0.00115
5.00	0.38593	-0.01993	0.01223	0.37613	-0.01378	0.00117	-0.00138	0.00064
5.50	0.37736	-0.01462	0.00915	0.37023	-0.01002	0.00065	-0.00076	0.00035
6.00	0.37109	-0.01065	0.00685	0.36596	-0.00720	0.00036	-0.00043	0.00020

6.50	0.36654	-0.00768	0.00512	0.36291	-0.00510	0.00020	-0.00024	0.00011
7.00	0.36329	-0.00546	0.00383	0.36077	-0.00353	0.00011	-0.00013	0.00006
7.50	0.36099	-0.00380	0.00286	0.35931	-0.00236	0.00006	-0.00007	0.00003
8.00	0.35942	-0.00255	0.00214	0.35836	-0.00149	0.00003	-0.00004	0.00002
8.50	0.35839	-0.00162	0.00160	0.35778	-0.00085	0.00002	-0.00002	0.00001
9.00	0.35776	-0.00093	0.00120	0.35747	-0.00040	0.00001	-0.00001	0.00000
9.50	0.35743	-0.00040	0.00091	0.35735	-0.00011	0.00000	-0.00001	0.00000
10.00	0.35733	-0.00000	0.00071	0.35733	0.00000	0.00000	-0.00000	0.00000